GopherMath

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Supporters

Generation Next, UMN President's Office, Campbell Family Foundation

GopherMath

GopherMath is a collaboration with Minneapolis Public Schools to support implementation of the Rational Number Project (RNP) curriculum developed at the UMN. It brings together the STEM Education Center, Department of Educational Psychology, and the Institute of Child Development.

This project builds on faculty expertise in four areas: fraction learning, teacher development, whole number assessments and interventions, and parent involvement. This is a unique opportunity for cross-department collaboration with children and parents from a high needs school.

Three Components

- Whole Number Intervention (Robin Codding, Michele Mazzocco, & Stacey Brandjord)
- Parent Engagement (Keisha Varma)
- Classroom Instruction and Teacher Professional Development (Erin Baldinger, Kathleen Cramer, & Chelsey Fagerlund)

School Site

The school serves a high immigrant population with approximately 80% of the families Somali.

The project builds on the school's current infrastructure to implement project activities:

- Monthly school parent meetings
- Prior experience with the RNP curricula
- Professional Learning Communities

Timeframe

- Whole number screening & intervention: October 2016 - February 2017
- Parent engagement: October 2016 – May 2017
- Classroom instruction with RNP:
  - Grade 3 January – February 2017
  - Grades 4 and 5 March – April 2017
- Teacher PD: October 2016 – April 2017

Whole Number Intervention

Strengthening whole number concepts is critical for developing fractional understanding. One approach to strengthening whole number knowledge is to ensure productive practice opportunities.

Productive practice opportunities are brief, occur daily, and ensure that students practice with materials that are an appropriate instructional match, indicated by highly accurate performance.

One way to faciliate productive practice opportunties in classrooms is to supplement core instruction with peer assisted learning strategies (PALS) by organizing activities in pairs or small groups of students.

The class-wide peer tutoring approach used in GopherMath involved students working in pairs on similar skills. Students were given examples of different strategies to solve basic facts, played a game demonstrating the conceptual basis of the strategy, and then took turns, as coach and player, presenting number facts to one another.



Data collection: Fall, Winter, Spring

- Number sense skills (magnitude comparison, number representation, decomposition)
- Grade level computation knowledge
- Fluency with basic facts

Planned data analysis

- Does basic number sense knowledge predict
  - computation knowledge following the peer tutoring intervention?
  - growth on basic fact fluency following the peer tutoring intervention, or resistance to intervention?
- Does number sense knowledge and fact fluency predict rational number knowledge?

Parent Engagement

The goal of the parent engagement program is to help parents better support math learning. As a part of this partnership, the school has invited us to their monthly parent meetings.

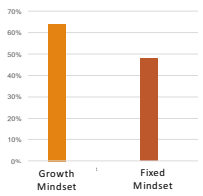
We measure the impact of these activities via parent and student survey data, student interviews, parent focus groups, and metrics from technologies used to support parent involvement (i.e., text messaging, YouTube videos).

Parent Engagement Meetings

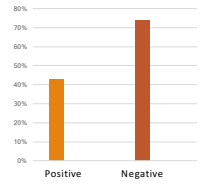
	Topic	Activities
1	Sharing ideas about math	Parent focus groups, Number talk, introduction to fraction circles
2	How your children are learning about fractions	Number talk, fraction circle distribution to families, fraction circle activity and discussion
3	Growth mindsets in math	Number talk, mindset survey and discussion
4	Connections between classroom instruction and math activities at home	Number talk, introduction to text messaging and YouTube videos
5	Promoting perseverance, grit, and growth mindsets	Number talk, parent focus groups

Preliminary insights

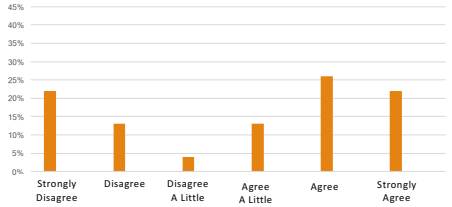
Percentage of parents with growth or fixed mindset



Percentage of parents who view mistakes as positive or negative



Percentage of parents who agree or disagree with the statement: "When I work hard, it makes me feel as though I'm not very smart."

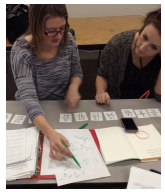


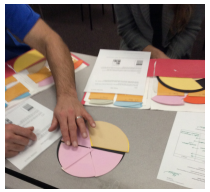
Teacher Professional Development

We provide professional development (PD) for the school's mathematics specialist and 6 teachers in grades 3-5 to optimize their use of RNP to support students' fraction learning. PD is woven into the school's current professional learning community meetings and involves:

- Investigating curricular materials
- Analyzing student thinking
- Developing parent support materials
- Unpacking big ideas underlying rational numbers

Data collection includes interviews with all participants at the beginning and end of the PD, focusing on their beliefs, understanding, and practices related to teaching about fractions.





Classroom Instruction

- Student performance data includes Grade 3 to 5 pretests and posttests, and interviews with 9-12 students per grade.
- Test data illustrates students' understanding of the state's fraction standards for each grade level.
- Interview data captures students' reasoning and conceptual understanding.
- Student data highlights the strengths of the curriculum and provides insights into how to adapt it to better support students' learning.
- Student performance data is woven into teachers' PD to help teachers see how to use student thinking to adjust future instruction.